MC 501 Data Wrangling in R rubric

50 points. Due Monday, November 16, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 50. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: the worksheet's code, 50 to 150 lines with its comments, with each of the three reviewer-facing comments 40 to 100 words. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- The worksheet is 11-wrangling-data.qmd. Fill every blank; the blanks are the steps the rows below grade. The Data Wrangling risks guide shows the traps on this job and how to check for each.
- Inferencing Data needs all four kinds of variable from this table: two categorical, one with three or more groups, and two continuous, plus your hand-coded variable. Part 4 is where they are kept.
- What it feeds: Describing Data and Inferencing Data both read `twitch_analysis.RDS`, and your hand-coded variable inside it is tested in Inferencing Data. Your White Paper reuses the moves in Part 5 (read a coding form in, make the codes factors, join by id) on your own data; the Twitch table itself stays here.
- Code the AURA Lab R Companion writes into your worksheet counts as your code. Keep the comment line it writes above each block, and be ready to explain every line.
- The comments the R Companion writes describe what the code does and do not count toward the three reviewer-facing comments.
- Before you submit: restart R and render the worksheet from the top, and see what breaks.

Data Wrangling in R, 50 points

Criterion	Exemplary	Proficient	Developing	Beginning

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1. Hand-code, and read your codes back 4 points	**4 points.** The worksheet's variable, reaction_only, hand-coded for all thirty messages Part 0 draws with `sample_messages()`, recorded in the coding form, read back with `read_csv()` and `clean_names()`, made a labeled factor with `factor()`, and joined to the analysis table by message id.	**3 points.** Hand-coded and read back, but the codes are not made a factor, or fewer than thirty messages carry a code after the join.	**2 points.** Codes typed into the worksheet instead of read from the coding form, or the thirty are not the messages Part 0 draws.	**0 to 1 points.** No hand-coding.
2. Load 4 points	**4 points.** The chat and stream tables loaded inside the worksheet.	**3 points.** Loaded from a file on your own computer, so the worksheet fails anywhere else.	**2 points.** Only one table loaded.	**0 to 1 points.** Not loaded in the worksheet.
3. Timestamps 4 points	**4 points.** Converted from epoch milliseconds to a date-time with `clean_dates()`, which divides by 1,000, or by hand, dividing by 1,000 first.	**3 points.** Converted, with a small error left, such as the time zone.	**2 points.** Converted without dividing by 1,000.	**0 to 1 points.** Not converted.
4. Message length 4 points	**4 points.** A new column holding each message's length.	**3 points.** Computed on the wrong field.	**2 points.** Computed but not saved as a column.	**0 to 1 points.** Missing.

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5. Join, and keep the analysis columns 4 points	**4 points.** Stream information joined on, with a comment naming the key, keeping the channel's main category (main_game), `is_gaming`, `viewer_band`, and one continuous audience measure (`median_viewers`, or each message's viewers from the snapshot nearest before it, which `join_chat_streams()` attaches).	**3 points.** Joined and the columns kept, but no comment names the key, or the continuous measure is missing.	**2 points.** The join matches nothing or multiplies the rows, unnoticed; or only `is_gaming` survives, so the table supports a t-test and nothing else.	**0 to 1 points.** No join.
6. Missing values 4 points	**4 points.** The count reported, and the decision stated.	**3 points.** Handled, but the count is not reported.	**2 points.** Dropped without saying how many.	**0 to 1 points.** Not handled.
7. Export 4 points	**4 points.** Saved as `twitch_analysis.RDS`.	**3 points.** Saved under another name.	**2 points.** Saved in a format that loses the column types.	**0 to 1 points.** Not exported.
8. Three reviewer-facing comments 15 points	**15 points.** Three comments, each naming a decision a reviewer could challenge, the rule you applied, and why.	**11 to 14 points.** Three comments, but one states the rule without the reason, or names a decision nobody would question.	**8 to 10 points.** One or two comments, or comments that describe what the code does instead of a decision you made.	**0 to 7 points.** None.
9. Clean-session run 7 points	**7 points.** Renders start to finish after restarting R.	**5 to 6 points.** Renders after one small fix, such as a missing `library()` line.	**4 points.** Stops partway in a fresh session.	**0 to 3 points.** Runs only with objects already in your environment, or not at all.